

BALOGUN, Rufai Omowunmi

Graduate School of Geography
Worcester, Massachusetts, 01610, USA

rufaibalogun1@gmail.com; rbalogun@clarku.edu

LinkedIn: <https://www.linkedin.com/in/rufaiomowunmibalogun/>

Website: <https://rufaibalogun.com/> GitHub: <https://github.com/Ruphai>

Orcid: [0000-0002-7741-2957](https://orcid.org/0000-0002-7741-2957)

EDUCATION

- 2028 (Expected) **PhD, Geography and Environmental Studies**, Clark University, USA
Major: Remote Sensing and Geospatial Analytics
First Minor: Machine Learning and Artificial Intelligence
Second Minor: Microwave Remote Sensing (Synthetic Aperture Radar and Radiometry)
- 2022 **MSc., Geo-Data Science**, University of South Brittany, France. (Distinction)
Thesis: The Transferability of Visual Language Pretrained Models to Remote Sensing Data
- 2021 **MSc., Earth Observation and Geo-informatics**, University of Salzburg, Austria. (Distinction)
- 2019 **B.Tech., Meteorology and Climate Science**, Federal University of Technology, (First Class Honors)
Thesis: Climate Variability and its implication on Vegetation Health in the Climatic Zones of Nigeria

PROFESSIONAL EXPERIENCE

- Sept. 2024 – present** **Research Assistant**, Center for Geospatial Analytics, Clark University, Massachusetts, United States of America (USA)
- **Explainable AI for Geospatial (Earth Observation) Foundation Models**
 - o Geospatial Analytics and Machine Learning application to downstream Earth Observation Processes
 - o Geospatial Foundation Model (Prithvi Global Model) Finetuning for ForestNet datasets,
 - o Explainability of Geospatial Foundation Model (Spectral, Spatial-Temporal and understanding the distribution of the latent space)
 - **AI Data Benchmarking**
 - o [GFM-Bench](#): A Benchmark to evaluate Geospatial Characteristics of Foundation Models (in preparation)
 - o Curating datasets for flooding, wildfire and landslide events for understanding the discriminative behavior of Geospatial Foundation Models in the Embedding space.
- Oct. 2022 – July 2024** **EO Data Science Researcher**, EURAC Research, Bolzano, Italy
- **Open data and open science:**
 - o Generated STAC Collection JSON for different datasets within the InterTwin project (CERRA, CERRA Land, EOBSv8, Soil Grid, and MERIT Hydro),
 - o Prepared Data Processing, Data Sharing, and Data Validation Exercises (using [OpenEO API](#)) to the [ESA MOOC: Cubes and Clouds: Cloud Native Open Data Sciences for Earth Observation](#) available on [EO College Platform](#),
 - o Co-led the writing and contributions to the “Alpine Drought Observatory: an operational drought monitoring platform” paper; submitted and currently under review by the Journal of Environmental Modelling and Software ([preprint](#))
 - **Open-Source Software development:**
 - o Provided an automated approach for gap filling [the Alpine Drought Observatory](#) meteorological datasets and drought indicators during in-house Cluster (Kubernetes) shutdown,
 - o Co-maintained the [Raster2STAC](#) python library for generating STAC Collections from Raster datasets (Cloud Optimized GeoTIFFs and KerChunk) within the [InterTwin](#) project,
 - o Developed and maintained the [SMODEX Python Package](#) for understanding the evolution of soil moisture and soil moisture anomalies using climate datasets, maintained the [PYSMM](#) python package for generation of soil moisture from SAR using Gradient Boosting Regression Techniques
 - **Data Analytics Pipeline Management and CI/CD:**
 - o Maintaining the data processing pipelines (CI/CD) and infrastructure for the Alpine Drought Observatory to ensure consistent time-series and daily production and deployment of drought indices on the [web portal](#),
 - o Soil moisture map generation from SAR using Google Earth Engine for the [Soil moisture Estimation optimized for Alpine areas \(SEA\)](#) project for United States Department of Agriculture, [Bird Watch](#) Project, South Tyrol and other Areas of Interests.

Sept. – Oct. 2022 Geospatial Analyst, CDL GeoHUM Lab, Salzburg, Austria (Contract)

- **Earth Observation/ Geospatial Data Analytics:**
 - o Curated a [population spatial indicator](#) and preliminary research for assessing the status of IDP camps for potential initiation of a Mass Drug Administration (MDA) for Malaria in the East African Rift (Uganda, DRC, Burundi, and Rwanda),
- **Geospatial Cloud Computing**
 - o Analyzed and proposed different cost setups for setting up an ArcGIS Image Server on the AWS Cloud Platform using services like the RDS, EC2 and Application Load Balancer for effective Image cataloguing and mosaicking.

Nov. 2021 – July 2022 Student Researcher, IRISA Group, University of South Brittany, France

- **Artificial Intelligence for Earth Observation**
 - o Researched and developed a methodological framework for evaluating the “*transferability of visual language models to remote sensing datasets with low versus high image resolutions*” using a visual question answering model framework,
 - o Developed and implemented a remote sensing visual question answering model using the elementwise and multi-modal tucker decomposition fusion strategy.
- **Project and team management**
 - o Served as a project manager for a team of 8 international students participating in the [On Cloud N: Cloud Cover Detection Challenge](#).

May 2020 – June 2022 Research Partner, The Sixth Axis, Ondo State, Nigeria

- **Users and community need assessment:**
 - o Co-developed the methodological framework and ocean expert database for User Consultation for assessing priority coastal hazards for expert users and local users across 15 West African countries,
- **Earth Observation/ Geospatial Analytics**
 - o Curated hydrometeorological datasets for flash flood and coastal flood vulnerability index computation for the [EARWAC Platform](#), presented at the Conference of Party (COP26), Glasgow, United Kingdom and the Living Planet Symposium, Bonn, Germany

July – Sept. 2021 Geospatial Analyst, Space4Good, The Hague, Netherlands (Internship)

- **Earth Observation/ Geospatial Analytics**
 - o Implemented a workflow [for individual tree-top identification](#) and extracted their corresponding crowns from Aerial Imageries using the one-stage object detector, RetinaNet (*Remote work coordinated within an agile project management framework*)
 - o Provided technical documentation and analysis of results to project stakeholders during sprint reviews.

Jan. – Aug. 2020 Geospatial Data Analyst, Center for Space Research and Applications, Akure (CESRA), Nigeria

- **Earth Observation/ Geospatial Analytics**
 - o Prepared [statistical analysis](#) (LOESS Regression, Correlation, Trend Testing – Mann Kendall and Sen’s slope, and anomaly detection) of climate and remote sensing data across 8 Nigerian cities,
 - o Contributed to the center’s annual training of university staffs and students on the use of GIS and Remote Sensing

June – Dec. 2018 Research Intern, National Space Research and Development Agency (NASRDA), Nigeria

- **Earth Observation/ Geospatial Analytics**
 - o First introduction to practical remote sensing working with different teams of researchers on different thematic areas including malaria mapping, climate anomalies, urban heat islands and flood mapping.
 - o Taught and supported other professionals in the monthly remote sensing and GIS training hosted at the Space Application Department

RESEARCH INTERESTS

Remote Sensing of Hydrometeorological Hazards,
Explainable Machine Learning and Geospatial Foundation Models,
Physics-Informed Machine Learning for Environmental Hazards,
Soil Moisture Properties Retrievals from Microwave Remote Sensors

SKILLSETS

Python Programming: Numpy, Pandas, Geospatial Libraries, xArray, Dask, PyTest

Remote Sensing/Geospatial Analysis: ArcGIS Products, QGIS, eCognition, Google Earth Engine, SNAP

Spatial Databases: SQL, Postgres/ PostGIS

Machine Learning: sklearn, pytorch, dvc, mlflow, wandb
Cloud/HPC: AWS, SLURM, GitLab CI, Linux/Bash, Docker/Kubernetes
Agile Project Management: G-Suite, Slack, Sprint Planning, Risk Management
Scientific Research and Documentation: LaTeX, Markdown/reStructuredText

CONTRIBUTED CONFERENCE PRESENTATIONS

- Hamed Alemohammad, Denys Godwin, **Rufai Balogun**, Sam Khallaghi, Sujit Roy, Rahul Ramachandran. [GFM-Bench: A Benchmark to Evaluate Geospatial Characteristics of Foundation Models](#). American Geophysical Union Fall Meeting 2024 (AGU24). Washington DC, December 2024.
- Zellner P. J., M Claus, T Dolezalova, **R. O. Balogun**, J Eberle, H Hodam, R Eckardt, S Meißl, A Jacob, A Angheloa. *MOOC Cubes and Clouds – Cloud Native Open Data Sciences for Earth Observation*. FOSS4G Europe, Tartu, Estonia. July 2024
- Zellner P. J., **Balogun R. O.**, Eckardt R., Hodam H., Dolezalova T., Meißl S., Eberle J., Claus M., Callegari M., Jacob A., Angheloa A. [Cubes & Clouds – A Massive Open Online Course for Cloud Native Open Data Sciences in Earth Observation](#). The European Geoscience Union (EGU), Vienna, Austria, April 2024
- **Rufai O. Balogun**, Peter J. Zellner, Felix Greifeneder. [SMODEX: a python package to understand the evolution of soil moisture anomalies](#). The South Tyrol Free Software Conference (SFSCON), NOI Tech Park, Bolzano, Italy, November 2023
- P. J. Zellner, **R. O. Balogun**, T. Iacopino, L. Cattani, M. H. Alasawedah, M. Claus, B. Ventura, A. Vianello, B. Giacomo, A. Jacob. [The Alpine Drought Observatory, how to organize and share data in a cross-national mountain region following the fair data principles](#). EuroGEO Workshop, Bolzano, Italy, October 2023
- P. J. Zellner, L. Cattani, G. Bertoldi, F. Greifeneder, M. H. Alasawedah, T. Iacopino, A. Vianello, **R. O. Balogun**, M. Claus, B. Ventura, K. Mayer, Z. Vlahovic, A. Jacob. [The Alpine Drought Observatory: An Alpine Wide Operational Drought Monitoring Platform](#). The Sixth Virtual Alpine Observatory Symposium, Grainau, Germany, March 2023
- A.D. Adebayo, **R.O. Balogun**, I. Adewunmi, S.O. Akande, O.C. Oghenechovwen, A. Giwa, [Enhancing Adaptation and Resilience against coastal multi-hazards along the West African Coasts \(EARWAC\)](#). ESA Living Planet Symposium, Bonn, Germany, May 2022
- **R.O. Balogun**, Ahmed Moussa, Margret Azuma, Meryeme Boumahdi, [Mapping burn severity using Sentinel-2 for 2020 California Wildfires](#). ISDE/GI Week, Salzburg, Austria, July 2021

PUBLICATIONS

- Zellner P.J., Claus M., Dolezalova T., **Balogun R.O.**, Eberle J., Hodam H., Eckardt R., Meißl S., Jacob A., Angheloa A. (2024). *MOOC Cubes and Clouds – Cloud Native Open Data Sciences for Earth Observation*. The International Archives of Photogrammetry, Remote Sensing and Spatial Information Sciences. XLVIII-4/W12-2024, 157–162. DOI: <https://doi.org/10.5194/isprs-archives-XLVIII-4-W12-2024-157-2024>
- Peter J. Zellner, **Rufai O. Balogun**, Konrad Mayer, Thomas Iacopino, Luca Cattani, Mohammad H. Alasawedah, Daniela Quintero, Michele Claus, Bartolomeo Ventura, Andrea Vianello, Alessio Salandin, Elisa Brussolo, Ziva Vlahovic, Christian Ronchi, Bertoldi Giacomo, Mariapina Castelli, Felix Greifeneder, Alexander Jacob (2023). *The Alpine Drought Observatory: an operational drought monitoring platform*, Environmental Modelling and Software (under review– preprint available [here](#))
- P. J. Zellner, **R.O. Balogun**, K Mayer, T Iacopino, L Cattani, M. H. Alasawedah, M Claus, B Ventura, A Vianello, G Bertoldi, F Greifeneder, A Jacob (2023). *GIS Success Stories – The Alpine Drought Observatory*. Geoinformation for crisis management, Innovation in Disaster Prevention, Iceland Liechtenstein Norway Grants
- O. C. Oghenechovwen, A.D. Adebayo, **R. O. Balogun**, S.O. Akande, I. J. Adewumi, Y. A. Giwa, & V. E. Umurhuru, S Hebden. (2022) *Enhancing Adaptation and Resilience against multi-hazards along West Africa's Coasts (EARWAC)*. The CEOS/CGMS Working Group on Climate and the World Meteorological Organizations Use Cases for Climate Monitoring from Space.

OPEN-ACCESS DATASETS

- **Balogun R.O.**, Zellner P.J., & Greifeneder, F. (2024). Soil Moisture Anomalies – ERA5_QM Version 2.0. EURAC Research. DOI: <https://doi.org/10.48784/ea665ca2-0ceb-11ed-86c5-02000a08f4e5>
- Zellner P.J., Dolezalova T., Claus M., **Balogun R.O.**, Eberle J., Hodam H., Eckardt R., Meißl S., Jacob A., and Angheloa A. (2024). *Data from Cubes & Clouds – Cloud Native Open Data Sciences for Earth Observation. Cubes and Clouds – Snow Cover STAC Collection*. DOI: <https://zenodo.org/records/10513915>

COURSES AND TUTORIALS

- **A Comprehensive Tutorial on Trajectory Analysis with R.** (Contributors: Balogun R.O., Harrison Leduc, Daniel Neau, Sunita Phuyal (2024)) URL: <https://github.com/HLed12/TrajectoryAnalysisTutorial?tab=readme-ov-file>
- **ESA MOOC: Cubes and Clouds – Cloud Native Open Data Science for Earth Observation.** URL: <https://eo-college.org/courses/cubes-and-clouds/>

LEADERSHIP AND COMMUNITY SERVICE

- Sept. 2022 – present** **Young Professionals Ambassador** (Industry), IEEE Geoscience and Remote Sensing Society (GRSS)
- Leading the *Chapter Industry Outreach Initiative* across all IEEE GRSS regions to activate local chapters involvement with industrial partners,
 - Leading the “[IEEE GRSS Industry event: Building your professional network in the geoscience industry](#)” outreach event at the European Geosciences Union since 2023,
 - Co-led the organization of the [IEEE GRSS Student and Young Professionals Summit \(virtual\)](#) reaching over 350 participants,
 - Co-developing young professional programs at the IGARSS conferences with other young professionals.
- July 2018 – Aug. 2019** **Project Mentor**, Care About Climate
- Served as a mentor to 10 young environmental professionals to help them implement their environmental sustainability projects in an effective way.

TECHNICAL REVIEW SERVICES

1. **Competitions/ Hackathons**
 - a. **Ready for Impact (RFI) Competition (2025)**, NASA Lifelines: *Evaluated about 60 project proposals and selected 13 finalists submitted for the NASA Lifelines Ready for Impact competition as part of a 4-member panel assessing its fitness for purpose, innovation, quality and sustainability, and engagement with end users.*
 - b. **3 Minute Thesis (2024)**, International Geoscience and Remote Sensing Symposium: *Evaluated 50 thesis proposals and selected 10 finalists to present at the International Geoscience and Remote Sensing Symposium in Athens, Greece as part of a three-member committee.*
 - c. **Geo-Pitch Competition (2023)**, International Geoscience and Remote Sensing Symposium: *Assess 40 project ideas and selected 10 finalists to present at the International Geoscience and Remote Sensing Symposium in Pasadena, California, USA as part of a 3-member review committee.*
 - d. **United Nations Youth Hackathon (2021)**, United Nations Major Group for Children and Youth + United Nations Statistical Division: *Selected proposals to participate in the UN virtual Youth Hackathon with finalists presenting at Youth Circles 2022, at the United Arabs Emirates.*
2. **Proposal Review**
 - a. **Chapter Industry Outreach (2024/2025)**, Geoscience and Remote Sensing Society: *Lead and review proposals for the Chapter Industry Outreach competition from GRSS chapters working on activating industry activities in their local or regional communities.*

ADVISORY SERVICES

- 2021 – July 2022** **Member**, United Nations Major Group for Children and Youth (UN MGCY) Science Policy, Data Working Group
- July 2018 – July 2019** **Advisory Council Member**, World Oceans Day

OTHER SERVICES

- **Co-organizer**, IEEE Geoscience and Remote Sensing Society Student and Young Professional Summit (Virtual), 2024
- **Convener**, Town Hall, Building your Professional network in the Geoscience industry, European Geoscience Union General Assembly, 2024
- **Co-convener**, Building your Professional network in the Geoscience industry (Networking event), European Geoscience Union General Assembly, 2023

SHORT INTENSIVE PROGRAMS/ SUMMER SCHOOLS

May – June 2023	High Performance and Disruptive Computing in Remote Sensing Summer School
April 2022	Domain Adaptation and Generalization, Artificial Intelligence Doctoral Academy
January 2022	Advanced Language Processing Winter School, LIG Labs and Naver Labs Europe
July 2021	PAISS Summer School, MIAI and PRAIRIE
June – July 2021	Eo4GEO International Summer School: Intelligent Earth Observation, UNEP-GRID and ZGIS
April 2021	ZGIS Spring School: Data Science Methods within Geoinformatics, ZGIS, Salzburg
February 2021	RUS Remote Training: Ship Detection with Sentinel-1 using SNAP S-1 toolbox,
December 2020	Copernicus MOOC. (Online), ESA
November 2020	Geospatial Applications for Disaster Risks Management MOOC. UN SPIDER, ISRO (Online)

CONFERENCES AND WORKSHOP PARTICIPATION

July 2024	International Geoscience and Remote Sensing Symposium (IGARSS), 2024, Athens, Greece
April 2024	European Geosciences Union General Assembly, 2024, Vienna, Austria
November 2023	The South Tyrol Free Software Conference (SFSCON), Bolzano, Italy
October 2023	EuroGEO/OEMC Workshop, 2023, Bolzano, Italy
July 2023	International Geoscience and Remote Sensing Symposium (IGARSS), 2023, Pasadena, California, USA
April 2023	European Geosciences Union General Assembly, 2023, Vienna, Austria
July 2022	GI Salzburg 2022, Salzburg, Austria
May 2022	ESA Living Planet Symposium, 2022, Bonn, Germany
July 2021	ISDE/ GI Week, 2021 Salzburg, Austria (Hybrid)
May 2021	Conflict Early Warning and Early Action Conference, 2021 (remote)

CERTIFICATIONS

August 2022	Google Project Management Professional Certificate
April 2021	SAR: Hazards, University of Alaska, Fairbanks (edX).

AWARDS AND HONORS

1. **Program Fellowship, NASA Lifeline (2025):** Selected through a competitive application process to co-curate NASA Lifelines programs such as [Ready for Impact](#), [Supper Clubs](#), and the [Earth Science Review Board](#) aimed at building a community that connects Earth Scientists with Humanitarians within the Program Work Stream in partnership with DevGlobal.
2. **Next Generation Tech Booster Scholarship (2024),** Udacity (Online): Selected as one of the top 500 out of 15,000 global pools of applicants to participate in the Udacity Agile Software Development Program.
3. **Open Research Award, EURAC (2023):** Selected as one of three institutes for exemplifying the best practices of Open Science through the Alpine Drought Observatory project, prevailing over other 17 institutes/ research center that applied. The Alpine Drought Observatory was co-designed by a team of 9 developers, researchers and technicians at the Institute for Earth Observation, EURAC Research.
4. **High Performance and Disruptive Computing in Remote Sensing (HDCRS) Travel Grant (2023):** Selected as one of ten grantees awarded \$1000 to participate in the High Performance and Distributed Computing Summer School in Reykjavik, Iceland.
5. **ESA Living Planet Symposium Travel Grant (2022):** Awarded a travel grant to present research outcome for the “Enhancing Adaptation and Resilience against coastal multi-hazards along West Africa’s Coasts (EARWAC) at the Living Planet Symposium.
6. **ESA-Future Earth Joint Program Grant (2020):** Awarded the ESA-Future Earth grant worth **25,000 Euros** to create a research demonstrator for the COP26 using ESA CCI datasets and other earth observation datasets for coastal hazards in West Africa (EARWAC) as a part of 7 early career scientists.
7. **Erasmus Mundus Joint Master’s Degree Scholarship (2020):** Awarded the prestigious Erasmus Mundus Joint Master’s Degree scholarship worth about **49,000 Euros** to pursue the Copernicus master’s in digital Earth at the University of Salzburg and the University of South Brittany for the Geo Data Science track.

8. **Departmental Prize -- Best Graduating Student in Meteorology and Climate Science (2019):** *Awarded the Departmental Prize for being the Graduating Student with the Highest Cumulative Grade Point Average (CGPA) in the Department of Meteorology and Climate Science, Federal University of Technology, Akure, Nigeria.*

PROFESSIONAL MEMBERSHIP

- European Geoscience Union
- IEEE Geoscience and Remote Sensing Society
- International Society for Photogrammetry and Remote Sensing